



Day 24: Relational Database Services (RDS)

[General Interview Questions]

◆ 1. How do you launch an RDS instance for a Java-based web app (e.g., Tomcat)?

Answer:

Via AWS Console or CLI with:

- DB engine selection (e.g., MySQL/MariaDB)
- Instance size, subnet, VPC
- Public access (optional)
- Inbound security group rule (port 3306)

◆ 2. What information does a Java app need to connect to an RDS DB?

Answer:

- **JDBC URL:** jdbc:mysql://<endpoint>:3306/dbname
- Username & password
- Appropriate DB driver

◆ 3. What is the benefit of using RDS over a self-hosted DB with Tomcat?

Answer:

- Automatic backups
- Managed failover (Multi-AZ)
- Easy scaling
- No patching or maintenance

◆ 4. How do you back up a full RDS database manually?

Answer:

Use mysqldump or take a **manual snapshot** via AWS Console/CLI.

◆ 5. What is the difference between full backup and snapshot in RDS?

Answer:

- **Full backup** (mysqldump): Dump of structure and data
- **Snapshot**: Storage-level snapshot of the entire DB state

◆ 6. What is the use of schema-only backup?

Answer:

To track **table structure, indexes, constraints** without data — ideal for versioning and Git-based deployment.

◆ 7. How do you automate backups in RDS?

Answer:

Enable **automated backup** during RDS creation, set **retention period** and **backup window**.

◆ 8. What's the best strategy to move a database from local to RDS?

Answer:

- Export full DB using mysqldump
- Create RDS instance
- Import dump using mysql client CLI

◆ 9. How do you test DB connection between Tomcat and RDS?

Answer:

Use JDBC connection from app config or manually use MySQL Workbench or CLI from EC2/Cloud9 with same network access.

◆ 10. What role does IAM play in RDS connectivity?

Answer:

IAM controls API access to RDS services but **does not manage DB-level user access** unless using IAM DB Authentication (MySQL/PostgreSQL only).

◆ 11. What is the best way to version-control schema changes in a team project?

Answer:

Use **schema-only SQL dumps** and store in **Git**; apply changes via CI/CD or migration tools (Liquibase/Flyway).

◆ 12. Can you restore only one table from an RDS snapshot?

Answer:

No. Snapshots are **all-or-nothing**; to restore a single table, restore snapshot to new DB and export/import the table.

◆ 13. What's the best backup strategy for production RDS?

Answer:

- Enable **automated backups**
- Take **manual snapshots** before major changes
- Use **cross-region backups** for DR
- Schedule regular **mysqldump exports** for logical backups

◆ 14. How can you reduce downtime during DB restore or migration?

Answer:

- Use **read replicas** and promote
- Restore to **new DB instance** and switch endpoint
- Use **DB proxy or app-level failover**

◆ 15. What's the risk of relying only on snapshots for backups?

Answer:

Snapshots are stored in AWS and can't be moved easily outside. For true portability or version control, use **logical dumps** (full/schema/data) with external storage.